Using AutoResearch to Determine Which Factors Best Determine Whether or Not a Minor League Hitter Will be Successful in MLB

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Abstract

This project uses AutoResearch to create a feature library that will most accurately determine whether or not a minor league hitter will have success in the big leagues when and if they reach that level. Predicting the success of a prospect has been a challenging task throughout baseball's lifespan, due to data that often lacks complexity or is altogether incomplete, as well as the intrinsic inconsistency of the development process as a whole. Starting with a baseline lasso classification model with a Youden's J index of 0.817, I ran three AutoResearch loops that eventually raised the model's performance to an 0.846 Youden's J index thanks to the identification of upper-minors power-shape features, level-to-level progression deltas, and contextual defensive stats as the most influential predictors of success. The loops also found that debut age and amateur type did not contribute additional signal beyond the other predictors that were identified.

1 Introduction

It has been almost a quarter century since the book *Moneyball* was published and the analytics revolution in baseball took off, but one area of the game that hasn't quite been cracked yet: accurate prediction of the future success of minor league prospects that have yet to make their MLB debut. Minor league data is often sparse and incomplete, at least relative to data collected at the Major League level, and player development trajectories tend to have high variance to begin with. Plus, factors such as organizational competence and player personality are hard to quantify, yet play a big role in prospects' development.

This problem of identifying prospects who will achieve big league success is further complicated by a class imbalance issue: in my dataset of 17,706 minor league hitters between 2010 and 2025, only 259 (1.46%) achieved sustained MLB success by my definition. Traditional metrics such as ROC AUC cannot reliably be used to evaluate classification success with this level of imbalance, which motivated my use of Youden's J index (sensitivity + specificity − 1), which gives equal weight to a model's ability to identify true successes and true failures.

The use of AutoResearch in this project eliminates the need for manual feature engineering that would be necessary using traditional machine learning methods. This way, human error and bias in choosing features could be minimized, and the AutoResearch loop would be able to cycle through numerous guesses and checks without constant intervention.

My research question is: what measureable factors from a hitter's minor league career can most strongly predict whether or not they'll achieve sustained success in Major League Baseball?

2 Methods

2.1 Dataset

My dataset consists of a few hundred player-level aggregated statistics for minor league hitters between the years 2010 and 2015, all pulled from the Fangraphs.com API or the MLB Stats API. The levels of the minor leagues included in the dataset are Triple-A, Double-A, High-A, Low-A, and Rookie, which each level bringing 48 hitting statistics to the table.

The outcome variable `success` is a binary "Yes" or "No" classifier. A player is determined to have been successful in MLB if their average fWAR (Fangraphs' Wins Above Replacement) per season is greater than or equal to one, or if their fWAR per 162 games is greater than 1.5 and they've played in at least five MLB seasons. Again, 259 of the 17,706 minor league hitters were labeled as successful.

2.2 Baseline Model

My baseline is a lasso logistic regression model with a tuned penalty of $= 0.00599$ and a mixture set to $= 1$, which was fit using the `tidymodels` package in R. The preprocessing recipe includes feature engineering step `step_indicate_na()` to flag whether or not a player played at a given level, `step_impute_median()` to impute NA values, `step_normalize()` to standardize each vector, `step_zv()` to remove variables with zero variance, and `step_downsample()` to address class imbalance issues. The baseline model reached a cross-validated Youden's J of 0.817.

2.3 AutoResearch Loop

The framework used to run the AutoResearch loop consists of four files: `prepare.R`, `model.R`, `run.R`, and `program.md`. `prepare.R` is held back from being edited by the AI agent; it is used to define how to evaluate the model using Youden's J index and how to log each result in `results.csv`. The agent is allowed to modify `model.R`, which defines the preprocessing recipe and the specifications for the lasso logistic regression model. `run.R` calls those first two scripts to be run by the agent, and provides further instructions for logging results. And finally, `program.md` provides the agent with its instructions, its constraints, and when it must stop the loop.

Using the Codex CLI agent, I ran three separate loops, adjusting `program.md` each time based on what I wanted the agent to do differently. The first loop (iterations 1–38 in `results.csv`) focused on discovering features by transforming and combining existing hitting statistics. The second loop (iterations 48–59) included the introduction of raw minor league fielding data to be used in conjunction with discovered features. The third loop (iterations 66–70) incorporated the manual ablation results (iterations 60–65) to focus on pruning different branches of exploration and the original raw stats, while also testing professional debut age and amateur type (high school, college, international) as potential features. The iterations not mentioned (iterations 39–47) consist of manual experiments that helped me make changes to `program.md` between the first and second loops.

The loop was instructed to stop after 10 consecutive iterations without an improvement in Youden's J by at least 0.001, with a hard cap of 75 total iterations per loop.

3 Experiments and Results

3.1 Metric Trajectory

3.2 Loop Findings

In the first loop, the largest gains in Youden's J came from the addition of upper minor leagues (Triple-A and Double-A) power-shape features, such as weighted ISO, HR/FB rate, FB%, and their product, combined with Double-A to Triple-A progression deltas. The one direction that at this point seemed fruitless was the replacement of `step_downsample()` with `step_smote()` to handle class imbalance, which actually dropped Youden's J below the baseline of 0.817.

The second loop oversaw the introduction and integration of minor league fielding data, which allowed for the construction of defensive spectrum features such as `upper_spectrum_score`,

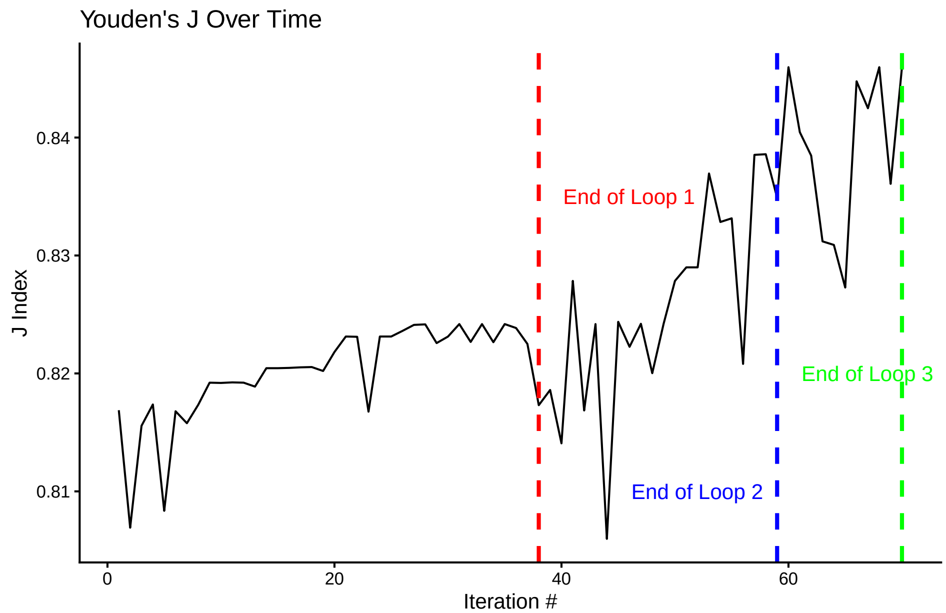

Figure 1: Youden's J index across all 70 iterations. Dashed veritcal lines indicate loop boundaries. Best result of 0.846 was achieved at iteration 60.

`upper_range_factor_9`, and position-weighted power interactions. I asked the agent to reconsider the use of `step_smote()` with the new data, where this time it improved performance over `step_downsample()`.

Finally, in the third loop the factors of debut age and amateur type were added to the dataset. The agent tested them both in their raw forms and in interaction with other existing features, but neither produced any meaningful uptick in Youden's J beyond what had already been captured in previous iterations.

3.3 Ablation Analysis

The ablation analysis I conducted following the conclusion of the second loop showed that removing upper minors weighted summary variables, while keeping their derived interaction terms, produced the eventual top result of 0.846.

Table 1 shows the contribution of each feature group to the best-performing model's performance.

Table 1: Ablation analysis table. Each row shows the Youden's J when the indicated feature group is dropped from the best model. Drop indicates the reduction from the second loop's best J of 0.839

Feature Group Removed	J index	Drop
None (full model)	0.839	0
Upper minors summaries	0.846	-0.007
Age-adjusted summaries	0.840	-0.001
AA to AAA deltas	0.838	0.000
Log-transformed variables	0.831	0.007
Lower minors deltas	0.831	0.008
Defensive variables	0.827	0.011

3.4 Best Model vs. Baseline

Table 2 summarizes performances across key model iterations.

Table 2: Performance comparison across key iterations.

Model	J index	Δ vs. Baseline
Lasso baseline	0.817	0
Best Loop 1 (iter. 31)	0.824	0.007
Best Loop 2 (iter. 58)	0.839	0.022
Best Loop 3 (iter. 68)	0.846	0.029
Best overall (iter. 60)	0.846	0.029

4 Discussion

My foremost finding throughout the loops and manual experiments is that the upper minors power-shape features, mostly ISO, HR/FB rate, and the product of the two weighted by plate appearances at Double-A and Triple-A, were the strongest predictors of success at the major league level among minor league hitters. This finding is consistent with traditional baseball thought: raw power that translates to the highest levels of the minor leagues, where the pitching is the best, is a strong signal for MLB readiness.

Another key finding is that a player's place on the defensive spectrum (the spectrum between the least demanding defensive positions and the most demanding ones) is also a high indicator of whether or not they'll find success at the big league level. Catchers and shortstops, who have the hardest jobs in the field, face a lower offensive threshold for MLB success than a first baseman or corner outfielder with identical numbers. My results suggest that the model captures this dynamic through `upper_spectrum_score` and its interactions with offensive features.

The fact that debut age and amateur type failed to produce meaningful upticks in Youden's J likely reflects that age-at-level variables already capture much of that same variance. Players who debut young, perhaps coming to the minors out of high school or an international signing, will appear younger relative to their level throughout their minor league career.

5 Failure Modes

The most common failure mode across the first two loops was the premature abandonment of exploration directions; the SMOTE method and lower minors signals were both dropped early on and not revisited by the agent in the first loop, despite providing a jump in Youden's J later on after I specifically asked the agent to re-introduce them.

In the second and third loops, the agent actually ignored my stopping criteria entirely, not persisting through my required 10 iterations without a 0.001 gain in Youden's J before stopping the loop. Instead, the agent would often circle around a singular exploration path that quickly proved unfruitful, and gave up early for whatever reason.

A third failure mode that proved critical during ablation analysis was the agent's inability to recognize redundant features, and its hesitance to explore the dropping of features when directed to do so. As the ablation table shows, the best model was actually achieved when a group of features was dropped from a model that included more advanced transformations and interactions of those features.

6 Limitations and Future Directions

The foremost limitation of this model is that it is limited to features and data than can be publicly accessed through the MLB Stats API and the Fangraphs.com API, which excludes potentially important variables. The addition of advanced batted ball data and defensive metrics, which are mostly held proprietary by teams throughout the minor leagues, could have provided further context to players' power output and defensive abilities.

Based on agent recommendations from all three loops, the most promising directions for future exploration may include the addition of draft round, draft position, and signing bonus to the dataset, as well as park-adjusted and league-adjusted minor league performance features.

A Best Performing Feature Library

The following two tables show all features present in the best performing model (iteration 60). Table 3 lists the 48 raw statistics that were recorded at each of the five minor league levels. Table 4 lists the intentionally engineered features constructed by the Codex agent.

Table 3: Raw statistics recorded at each of five minor league levels (R, A, A+, AA, AAA). Each stat appears five times in the feature set with the corresponding level suffix.

Feature	Description
Counting Stats	
g	Games played
ab	At bats
pa	Plate appearances
h	Hits
x1b	Singles
x2b	Doubles
x3b	Triples
hr	Home runs
r	Runs scored
rbi	Runs batted in
bb	Walks
ibb	Intentional walks
so	Strikeouts
hbp	Hit by pitch
sf	Sacrifice flies
sh	Sacrifice hits
gdp	Grounded into double play
sb	Stolen bases
cs	Caught stealing
balls	Total pitches taken as balls
strikes	Total pitches taken as strikes
pitches	Total pitches seen
Rate Stats	
avg	Batting average
obp	On-base percentage
slg	Slugging percentage
ops	On-base plus slugging
iso	Isolated power (SLG − AVG)
babip	Batting average on balls in play
bb_percent	Walk rate
k_percent	Strikeout rate
bb_k	Walk-to-strikeout ratio
sw_str_percent	Swinging strike rate
spd	Speed score
Batted Ball Profile	
gb_percent	Ground ball rate
fb_percent	Fly ball rate
ld_percent	Line drive rate
iffb_percent	Infield fly ball rate
hr_fb	Home run to fly ball rate
gb_fb	Ground ball to fly ball ratio
Spray Direction	
pull_percent	Pull rate
cent_percent	Center rate
oppo_percent	Opposite field rate
Weighted Stats (FanGraphs)	
w_rc	Weighted runs created (wRC)
w_rc_2	Weighted runs created plus (wRC+)
w_raa	Weighted runs above average (wRAA)
w_oba	Weighted on-base average (wOBA)
w_bs_r	Weighted base running runs
Other	
min_age	Player's age at that level

Table 4: Engineered features in the best-performing model (iteration 60).

Feature	Description
AA-to-AAA Progression Deltas	
`aa_aaa_hr_fb_delta`	Change in HR/FB rate from AA to AAA
`aa_aaa_fb_delta`	Change in fly ball rate from AA to AAA
`aa_aaa_power_shape_delta`	Change in ISO $\times$ FB% from AA to AAA
`aa_aaa_bb_minus_k_delta`	Change in BB% $-$ K% from AA to AAA
`aa_aaa_wrc_per_age_delta`	Change in wRC+ per age from AA to AAA
`aa_aaa_ops_per_age_delta`	Change in OPS per age from AA to AAA
`aa_aaa_wrc_delta`	Change in wRC+ from AA to AAA
`aa_aaa_iso_delta`	Change in ISO from AA to AAA
`aa_aaa_bb_delta`	Change in BB% from AA to AAA
`aa_aaa_k_delta`	Change in K% from AA to AAA
`aa_aaa_age_delta`	Change in age from AA to AAA
`aa_aaa_pa_delta`	Change in plate appearances from AA to AAA
`aa_aaa_ops_delta`	Change in OPS from AA to AAA
Log Transformations (Upper Minors)	
`log_iso_aa`	Log-transformed ISO at AA
`log_iso_aaa`	Log-transformed ISO at AAA
`log_hr_aa`	Log-transformed HR at AA
`log_hr_aaa`	Log-transformed HR at AAA
`log_fb_aa`	Log-transformed FB% at AA
`log_fb_aaa`	Log-transformed FB% at AAA
Low-A to High-A Progression Deltas	
`la_ha_iso_delta`	Change in ISO from Low-A to High-A
`la_ha_hr_fb_delta`	Change in HR/FB rate from Low-A to High-A
`la_ha_fb_delta`	Change in FB% from Low-A to High-A
`la_ha_power_shape_delta`	Change in ISO $\times$ FB% from Low-A to High-A
`la_ha_bb_minus_k_delta`	Change in BB% $-$ K% from Low-A to High-A
`la_ha_wrc_delta`	Change in wRC+ from Low-A to High-A
`la_ha_ops_delta`	Change in OPS from Low-A to High-A
`la_ha_wrc_per_age_delta`	Change in wRC+ per age from Low-A to High-A
Rookie to Low-A Progression Deltas	
`r_la_iso_delta`	Change in ISO from Rookie to Low-A
`r_la_hr_fb_delta`	Change in HR/FB rate from Rookie to Low-A
`r_la_fb_delta`	Change in FB% from Rookie to Low-A
`r_la_power_shape_delta`	Change in ISO $\times$ FB% from Rookie to Low-A
`r_la_bb_minus_k_delta`	Change in BB% $-$ K% from Rookie to Low-A
`r_la_wrc_delta`	Change in wRC+ from Rookie to Low-A
`r_la_ops_delta`	Change in OPS from Rookie to Low-A
`r_la_wrc_per_age_delta`	Change in wRC+ per age from Rookie to Low-A
High-A to AA Progression Deltas	
`ha_aa_iso_delta`	Change in ISO from High-A to AA
`ha_aa_hr_fb_delta`	Change in HR/FB rate from High-A to AA
`ha_aa_fb_delta`	Change in FB% from High-A to AA
`ha_aa_power_shape_delta`	Change in ISO $\times$ FB% from High-A to AA
`ha_aa_bb_minus_k_delta`	Change in BB% $-$ K% from High-A to AA
`ha_aa_wrc_delta`	Change in wRC+ from High-A to AA
`ha_aa_ops_delta`	Change in OPS from High-A to AA
`ha_aa_wrc_per_age_delta`	Change in wRC+ per age from High-A to AA
Defensive Spectrum Features (Upper Minors)	
`upper_def_inn`	Total defensive innings at AA and AAA
`upper_premium_inn`	Innings at premium positions (C, SS, CF)
`upper_corner_inn`	Innings at corner positions (1B, LF, RF)
`upper_infield_inn`	Innings at infield positions (2B, 3B, SS)
`upper_outfield_inn`	Innings at outfield positions (LF, CF, RF)
`upper_premium_share`	Share of innings at premium positions
`upper_corner_share`	Share of innings at corner positions
`upper_infield_share`	Share of innings at infield positions
`upper_outfield_share`	Share of innings at outfield positions
`upper_spectrum_score`	Weighted defensive spectrum score
`upper_chances`	Total defensive chances at AA and AAA
`upper_range_factor_9`	Range factor per 9 innings at upper minors
`upper_power_spectrum`	Power-shape $\times$ spectrum score interaction
`upper_control_spectrum`	BB% $-$ K% $\times$ spectrum score interaction
`upper_wrc_premium`	wRC+ $\times$ premium position share interaction
`upper_power_corner`	Power-shape $\times$ corner position share interaction